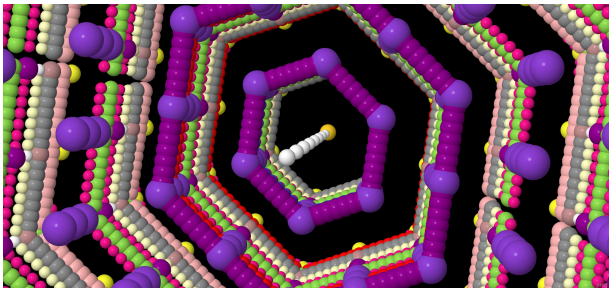
Representation and Construction of Perfects

An Illustrative Exposition of Fermatś Last Theorem and Bealś Conjecture



Dempster Welling

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Abstract

Having two or more identical sources with which to form the perfect constrains construction to two or more layers at a time which results in the exclusion of the apex.

1 Problems

$$S \begin{cases} A^2 + B^2 = C^2 & \text{Pythagorean Theorem} \\ A^N + B^N = C^N & \text{Fermat's Last Theorem} \\ A^X + B^Y = C^Z & \text{Beal's Conjecture} \end{cases}$$

2 Computational Algorithm

$$H_\Delta \begin{cases} 0 & \theta \text{ zero} \\ \Delta(\theta, \beta, \delta) & \theta, \beta \text{ and } \delta \text{ whole} \end{cases}$$

$$\Delta(\theta, \beta, \delta) = (\beta + \delta)\Delta((\theta - 1), \beta, \delta) + \delta \sum_{r=1}^{\beta-1} \Delta((\theta - 1), r, \delta)$$

$$\begin{aligned} C_\theta &= P^\theta - P^{(\theta-2)} \\ C_\theta &= P^{(\theta-2)}(P^2 - 1) \\ C_\theta &= P^{(\theta-2)}(P - 1)(P + 1) \\ C_\theta &= P^{(\theta-3)}(P - 1)(P)(P + 1) \end{aligned}$$

$$C_\theta \text{ mod } 6 \equiv 0$$

$$C_\theta \quad \text{has six identical parts when } \theta > 2$$

$$\begin{aligned} L0_\beta &= \Delta(\theta, \beta + 0, \delta = 1) - \Delta(\theta - 2, \beta + 0, \delta = 1) \\ L1_\beta &= \Delta(\theta, \beta + 1, \delta = 1) - \Delta(\theta - 2, \beta + 1, \delta = 1) \\ L2_\beta &= \Delta(\theta, \beta + 2, \delta = 1) - \Delta(\theta - 2, \beta + 2, \delta = 1) \end{aligned}$$

R is the differences in adjacent layers which characterize the ring

$$\begin{aligned} R1_\beta &= (L1_\beta - L0_\beta) \\ R2_\beta &= (L2_\beta - L1_\beta) \end{aligned}$$

$$\epsilon_\psi = (R2_\beta - R1_\beta)$$

$$\epsilon_\theta = \begin{cases} 1 & \theta = 2(flat) \\ \epsilon_\psi/6 & \theta > 2 \end{cases}$$

Equations shown here define the $H_p^y e$ mechanism. See Tables 3 6 7.

The mechanism generates the complete set of differences for perfects of all orders. The difference of two perfects is an additive method. The recursive function uses only addition. Perfects are formed from the subpolynomial clusters, C_θ. The cluster is shown to have a 6 symmetry(6s). The epsilon weights, ϵ_ψ are 6s characteristics of a layer which define the weight of the rings in the cluster. ϵ_ψ is the weight of single strata in a ring of the layer.

All closed form polynominal equations for ϵ_ψ are derived from the table. Tables 1 2 describes the growth of ϵ_ψ. The table has two sides. The $(\beta - 1)$ side has an alternating series of terms. The $(\beta + 1)$ side has an increasing series of terms. The rows in table are the coefficients of the polynomial sum. Alternating columns cancel out.

0	1	0											
1	1	1											
2	1	2	0										
3	1	3	2										
4	1	4	5	2									
5	1	5	9	7	2								
6	1	6	14	16	9	2							
7	1	7	20	30	25	11	2						
8	1	8	27	50	55	36	13	2					
9	1	9	35	77	105	91	49	15	2				
10	1	10	44	112	182	196	140	64	17	2			
11	1	11	54	156	294	378	336	204	81	19	2		
12	1	12	65	210	450	672	714	540	285	100	21	2	
13	1	13	77	275	660	1122	1386	1254	825	385	121	23	2

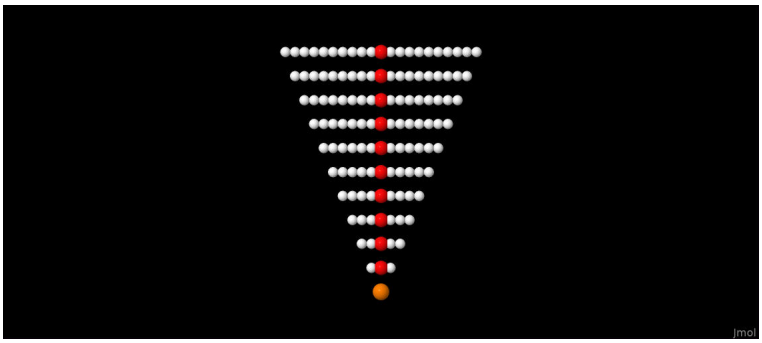
Table 1: $\Delta\ \epsilon$ growth L2

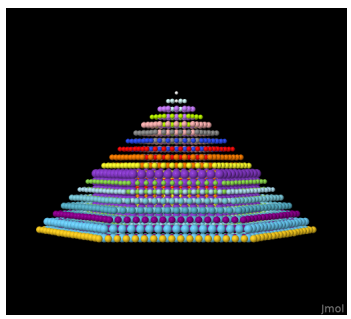
0	1	0											
1	1	-1											
2	1	-2	0										
3	1	-3	2										
4	1	-4	5	-2									
5	1	-5	9	-7	2								
6	1	-6	14	-16	9	-2							
7	1	-7	20	-30	25	-11	2						
8	1	-8	27	-50	55	-36	13	-2					
9	1	-9	35	-77	105	-91	49	-15	2				
10	1	-10	44	-112	182	-196	140	-64	17	-2			
11	1	-11	54	-156	294	-378	336	-204	81	-19	2		
12	1	-12	65	-210	450	-672	714	-540	285	-100	21	-2	
13	1	-13	77	-275	660	-1122	1386	-1254	825	-385	121	-23	2

Table 2: $\Delta\ \epsilon$ growth L0

$$\epsilon_3 = 6\beta$$
$$\epsilon_4 = 12\beta^2$$
$$\epsilon_5 = 20\beta^3 + 4\beta$$
$$\epsilon_6 = 30\beta^4 + 18\beta^2$$
$$\epsilon_7 = 42\beta^5 + 50\beta^3 + 4\beta$$
$$\epsilon_8 = 56\beta^6 + 110\beta^4 + 26\beta^2$$

3 Representation and Construction





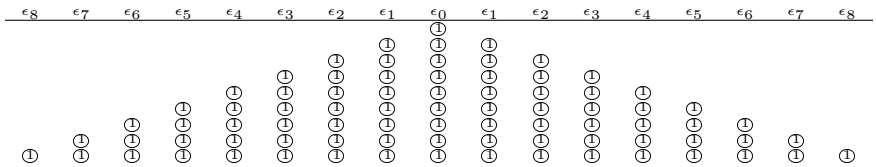
The perfect Δ^2 can be represented as a triangular arrangement of points. The perfect Δ^3 $\Delta^4 \ldots$ can be modeled in a hexagonal pyramid arrangement of points. The dimension or height of the perfect is the count of points in the center along with the apex; the ϵ_0 column. The number of layers is the count of points in the C_1 center.

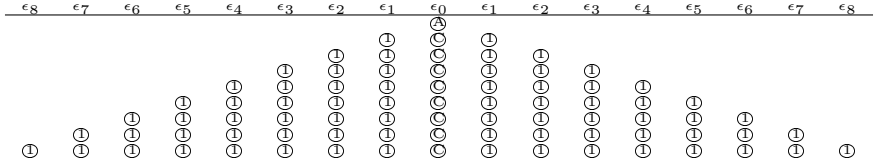
The 'volume' of a perfect is the summation of row elements, ϵ^θ, in the layer weight table. 3

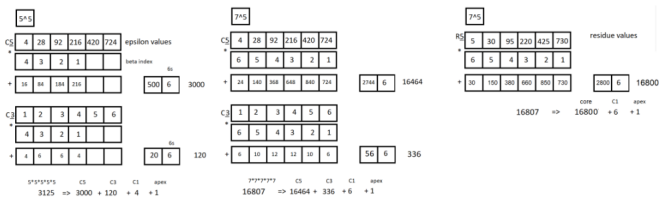
$$V(\Delta^\theta_\beta) = \sum_{x=1}^{\beta} LW(\theta, x)$$

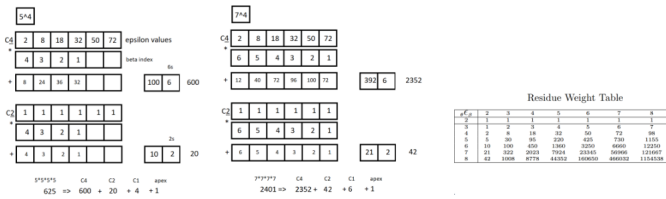
The 'volume' of the 3p wedge is the summation of row elements, ϵ^θ, in the residue weight table. 7

$$V(C^\theta_\beta) = \sum_{x=1}^{\beta-1}((\beta - x)RW(\theta, x))$$



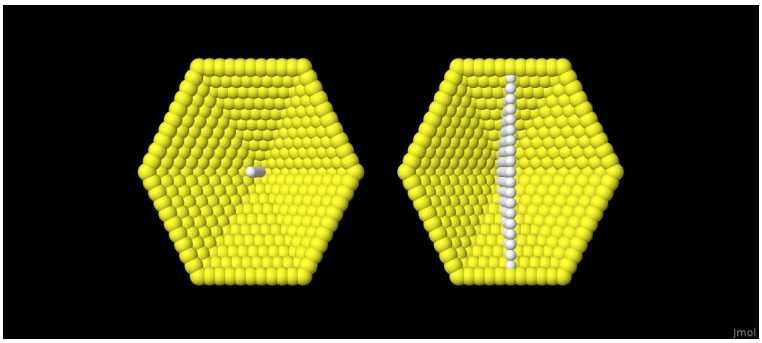


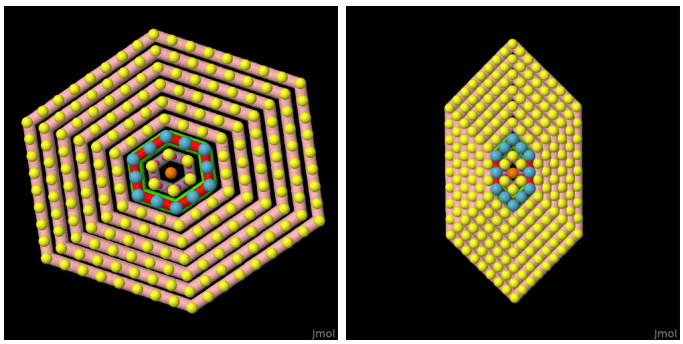




Residue Weight Table

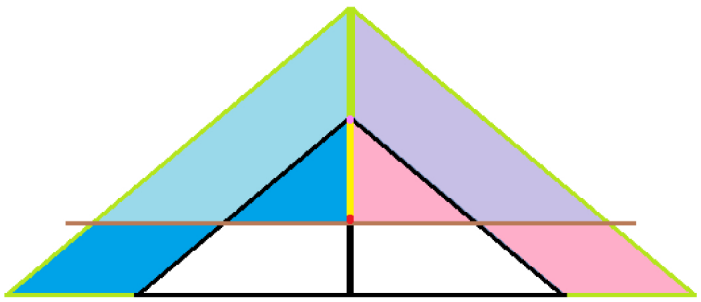
ϵ^n β^n	2	3	4	5	6	7	8	9
2	1	1	1	1	1	1	1	1
3	1	2	3	4	5	6	7	8
4	2	8	18	32	50	72	98	128
5	5	30	95	220	425	730	1155	1720
6	10	100	450	1360	3250	6660	12250	20800
7	21	322	2023	7924	23345	56966	121667	235368
8	42	1008	8778	44352	160650	466032	1154538	2542848

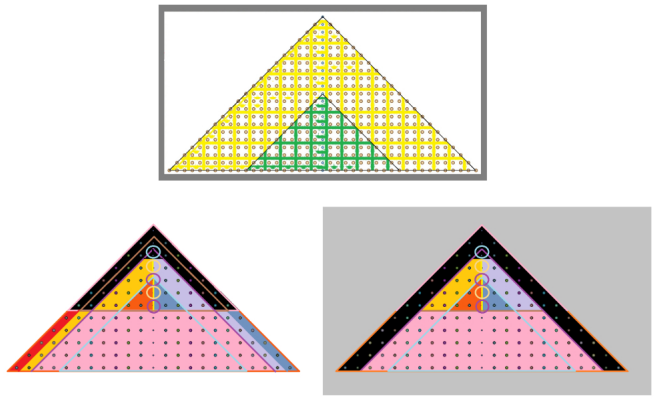




An odd perfect Δ^O has axial symmetry around the C_1 center. An even perfect Δ^E has a plane of symmetry about the C_2 center. The C_2 cluster is 2s around the C_1 center for bilateral symmetry. The odd orders are transformed into the same bilateral representation. Each 6s 3d wedge is represented as two 3p ϵ units around the C_1 center.

From the diagrams it can be seen that the difference operation is two sided and produces two symmetric parts which are avialable for rearrangement into the perfect $\Delta_A = \Delta_C$ - Δ_B.





There are three obstacles one must hurdle in order to rearrange points after the difference operation. First the 6s3p left and right core pieces are formed from 3p components, then the center must be formed from 1p units and finally an apex placed.

One stipulates to the first two of these conditions. Constructing the center requires that ϵ^{θ} units are decomposed into 1p units. The constraint of left and right sources force the center to be even. The center is formed from points in pairs. The newly forming object must be built two layers at a time. A 3p6s core can be formed for some measure of Δ^B and Δ^C. An even center may be placed, however infrequent. But finally to place the apex is a challenge. The apex can not be placed atop the ϵ_0 column.
A 2s3p core plus even constrained center plus apex results in an odd number of components.

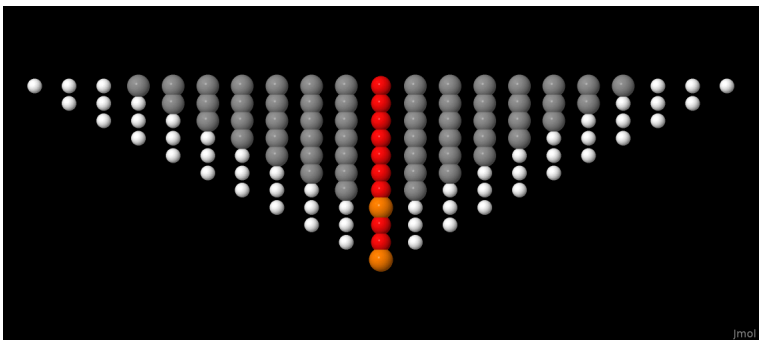
$$\Delta^{odd} = 1 + C_1 + C_3 + C_5 + \dots$$
$$\Delta^{even} = 1 + C_2 + C_4 + C_6 + \dots$$

$$\Delta^{odd} = apex + center + C_3 + C_5 + \dots$$
$$\Delta^{even} = apex + center + twotriangles + C_4 + C_6 + \dots$$

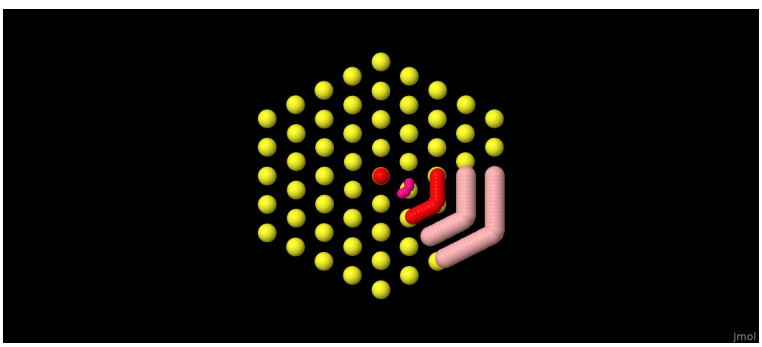
The difference operation[1] removes the apexs from both objects, ΔC and ΔB, prior to making the divet. A fulcrum is left in the divet. The missing divet is a constructable apex-less Δ_D similiar but smaller then ΔB. The divet is replenished with 3p points from the objects outer edges and/or c2 center difference

[1] in FLT case where $C^N - B^N = A^N$ and $A < B < C$

for even order. Some 3p points are fractured to fill the 1p center(two layers at a time). With divet and center complete the object forms an apex-less perfect. $\Delta_C - \Delta_B = \Delta_A - 1$. At this point Fermat's Last Theorem is shown by inspection.



The corner wedge arrangement is not bi-laterally symmetric but does maintain balance, there is a hex-cubic form, which when removed forms a divet in the object.



It is not possible to formulate a difference for all variables of $C^Z - B^Y = A^X$. It is possible to perfom an intersection of two objects. This leads to 'negative' space. To satisfy Beal's conjecture one must combine two objects.

Given the two objects, Δ_A and Δ_B form Δ_C. Steps to construct perfect; Seperate the apex and c1 centers from both. Make two identical piles left and right from clusters of A and B. Make a minimal center.

$$C1_{init} = (a + b) \mod 6$$
$$R_{frac} = (a + b) - C1_{init}$$

Distribute R_{frac} equally into the left 3p and right 3p piles.

When $C1_{init}$ is 1,2,3,4 or 5, one proceeds by forming the perfect Δ_C^X where X is the arbitrary order. When $C1_{init}$ is zero, one forms perfect after seeding with one of the apexes.

Seeded with 1, Δ_2^X, Δ_3^X, Δ_4^X, Δ_5^X or Δ_6^X construction proceeds adding two layers at a time until piles exhausted. Every layer added to the growing form completes the object, a larger perfect. For any value of order X, the same holds true.

Construction of a perfect is not exact when two identical processes are used to build object. The two processes, one for each pile, perform after the initial configuration is seeded. There is always an even number of components to be utilized after seeding. The left and right piles, two sided. The extra 1p apex is not used. $\Delta_A + \Delta_B = \Delta_C + 1$. The chances of forming a 2s3p core is very small, filling the center even smaller. One succeeds in forming the perfect when piles are exhausted and the final two layers complete but fails to use all the points.

4 Summary

The question, 'How close does one get when reforming perfects?', has a minimum of one.

$\Delta_A + \Delta_B = \Delta_C + 1$. When adding perfects together, an extra apex.

$\Delta_C - \Delta_B = \Delta_A - 1$. When performing a difference, a missing apex.

ψ	1	2	3	4	5	6	7	8	9
1	1	1	1	1	1	1	1	1	1
2	1	3	5	7	9	11	13	15	17
3	1	7	19	37	61	91	127	169	217
4	1	15	65	175	369	671	1105	1695	2465
5	1	31	211	781	2101	4651	9031	15961	26281
6	1	63	665	3367	11529	31031	70993	144495	269297
7	1	127	2059	14197	61741	201811	543607	1273609	2685817
8	1	255	6305	58975	325089	1288991	4085185	11012415	26269505

Table 3: Layer Weights the difference of two perfects $\Delta^{\psi}_{N+1} - \Delta^{\psi}_{N}$

ψ 2	2	3	4	5	6	7	8	9	10
1	2	2	2	2	2	2	2	2	2
2	4	8	12	16	20	24	28	32	36
3	8	26	56	98	152	218	296	386	488
4	16	80	240	544	1040	1776	2800	4160	5904
5	32	242	992	2882	6752	13682	24992	42242	672325
6	64	728	4032	14896	42560	102024	215488	413792	7378566
7	128	2186	16256	75938	263552	745418	1817216	3959426	79028487
8	256	6560	65280	384064	1614080	5374176	15097600	37281920	832227848

Table 4: Layer Weights the difference of two perfects $\Delta^{\psi}_{N+2} - \Delta^{\psi}_{N}$

ψ 3	3	4	5	6	7	8	9	10	11
1	3	3	3	3	3	3	3	3	3
2	9	15	21	27	33	39	45	51	57
3	27	63	117	189	279	387	513	657	819
4	81	255	609	1215	2145	3471	5265	7599	10545
5	243	1023	3093	7533	15783	29643	51273	83193	128283
6	729	4095	15561	45927	113553	246519	484785	882351	1509417
7	2187	16383	77997	277749	807159	2019027	4503033	9176457	17390019
8	6561	65535	390369	1673055	5699265	16386591	41367105	94235199	197581665

Table 5: Layer Weights the difference of two perfects $\Delta^{\psi}_{N+3} - \Delta^{\psi}_{N}$

ϵ_ψ	2	3	4	5	6	7	8	9
2	1	1	1	1	1	1	1	1
3	1	2	3	4	5	6	7	8
4	2	8	18	32	50	72	98	128
5	4	28	92	216	420	724	1148	1712
6	8	92	432	1328	3200	6588	12152	20672
7	16	292	1928	7704	22920	56236	120512	233648
8	32	908	8328	42992	157400	459372	1142288	2522048

Table 6: ϵ weights single strata weight units for rings

ϵ_ψ	2	3	4	5	6	7	8	9
2	1	1	1	1	1	1	1	1
3	1	2	3	4	5	6	7	8
4	2	8	18	32	50	72	98	128
5	5	30	95	220	425	730	1155	1720
6	10	100	450	1360	3250	6660	12250	20800
7	21	322	2023	7924	23345	56966	121667	235368
8	42	1008	8778	44352	160650	466032	1154538	2542848

Table 7: Residue Weight : sum of ϵ weights for all strata that form cluster